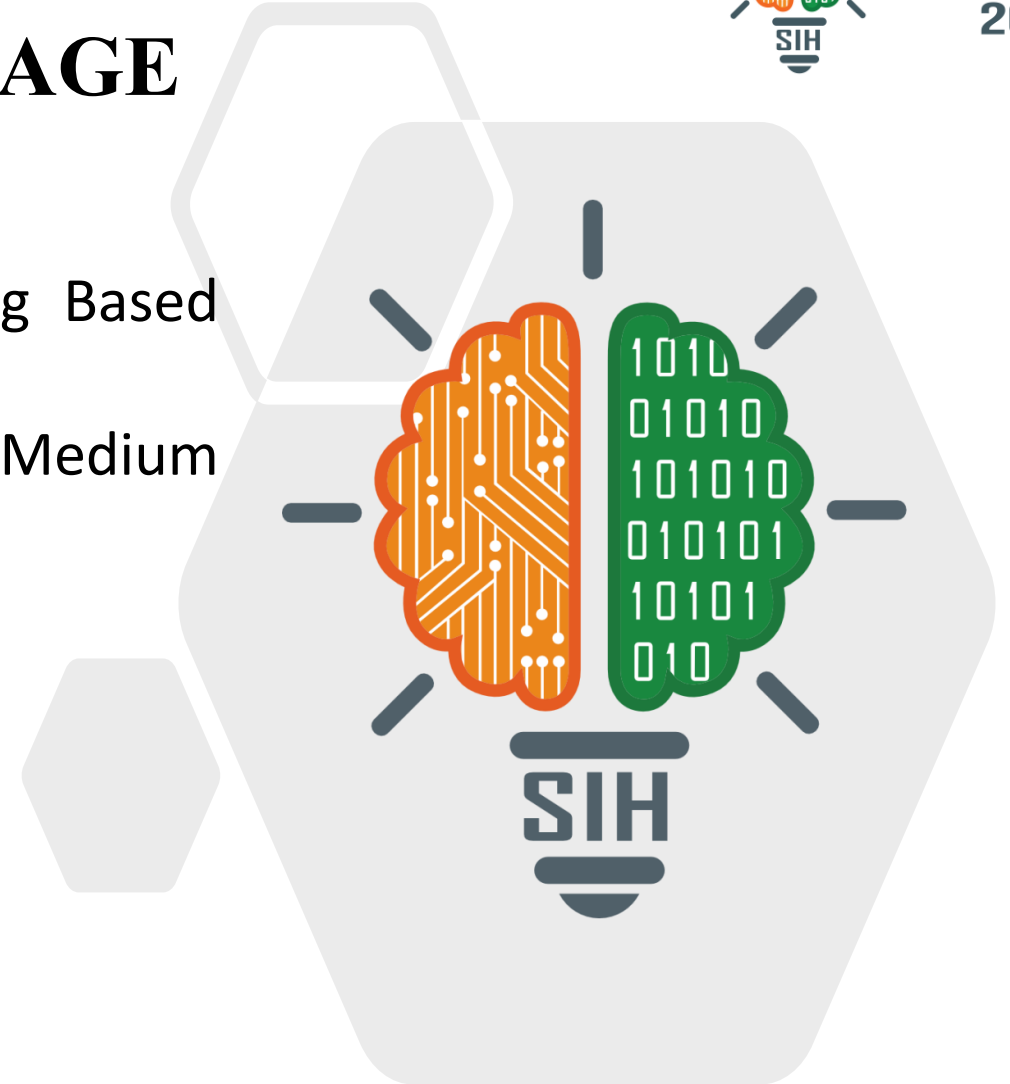


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID :** SIH26142
- **Problem Statement Title-** Deep Learning Based Super Resolution Mapping (SRM) from Medium Resolution Satellite Imageries
- **Theme-** Space Technology
- **PS Category-** Software
- **Team ID-**
- **Team Name -** Antariksh Setu



Resolvance- Sovereign Super-Resolution for Sentinel-2 10m→3.3m N-Channel with Spectral Truth & Uncertainty

Problem:

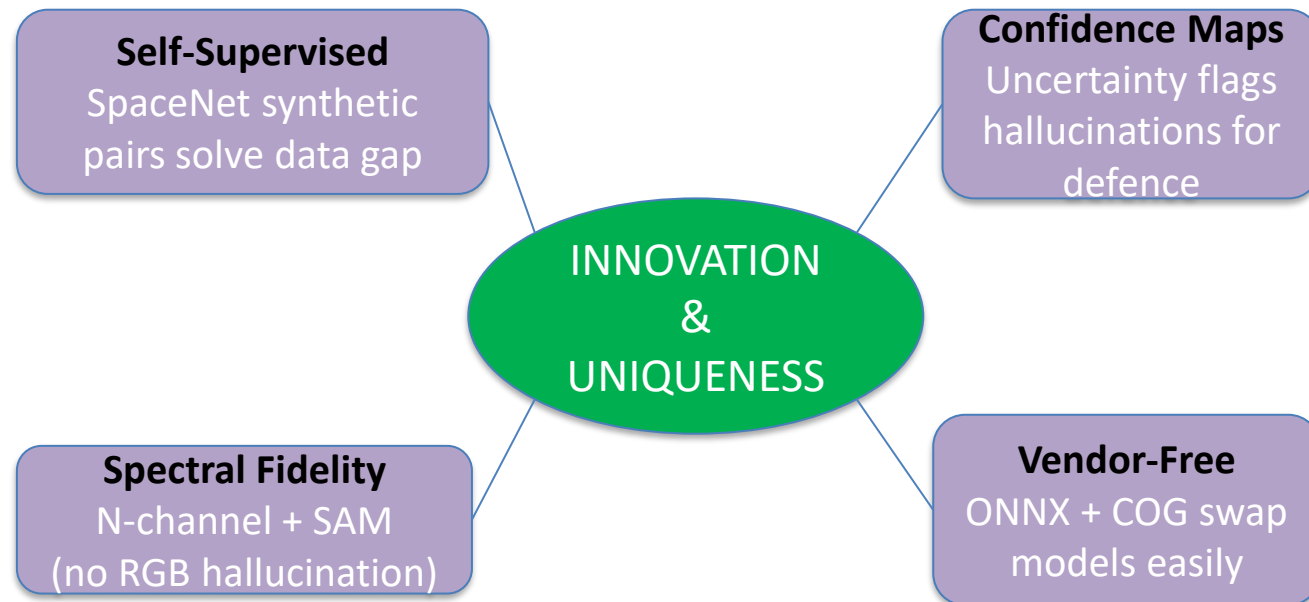
Free Sentinel-2 covers India every 5 days at 10m, but 10m hides roads, buildings and field edges and <1m commercial data costs \$1000s. Without 10m→<4m super-resolution, analysts work blind or overspend.

Our Idea:

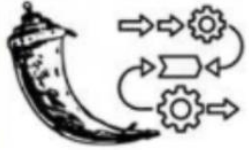
Resolvance is an AI-powered pipeline with a three-layer approach - N-channel SR, spectral SAM guard, and uncertainty heatmap. Ensuring reliable <4m mapping when 10m Sentinel-2 falls short. It uses tiled ESRGAN/SwinIR 3× super-resolution trained on synthetic SpaceNet pairs to deliver sharp, spectral-true, trust-scored COGs for any Indian terrain.

How It Addresses Problem:

- ✓ Eliminates \$1000s commercial buy — free 5-day revisit → <4m
- ✓ Enables on-premises processing with 2.6s CPU inference , no foreign API
- ✓ Provides tactical decision-making visibility on narrow roads/buildings/fields — QGIS COG



PIPELINE & EXTENSION



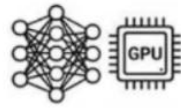
Python flask,
Rasterio/GDAL/Affine
(CRS + Affine
preservation)

GEOSPATIAL & DEMO



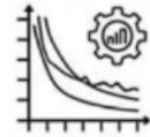
tiffle + Geopandas
+ QGIS validation,
React+Leaflet viewer
(Kepler fork)

INFERENCE



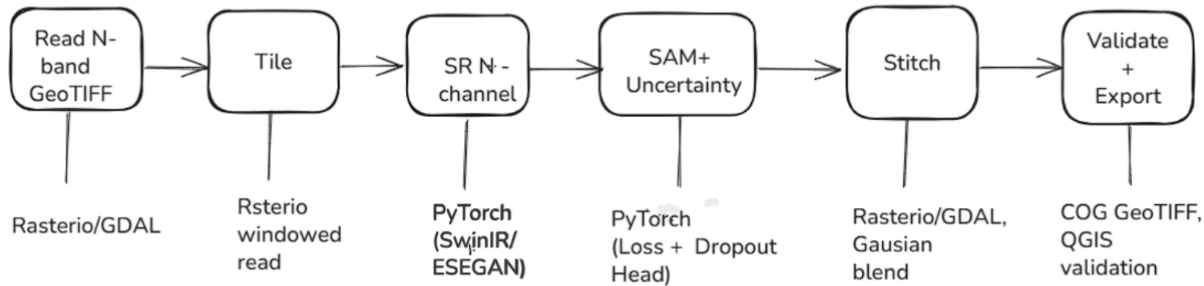
PyTorch + ESRGAN/SwinIR
(1.2MB-15MB models),
mixed precision inference

TRAINING, LOSS & METRICS

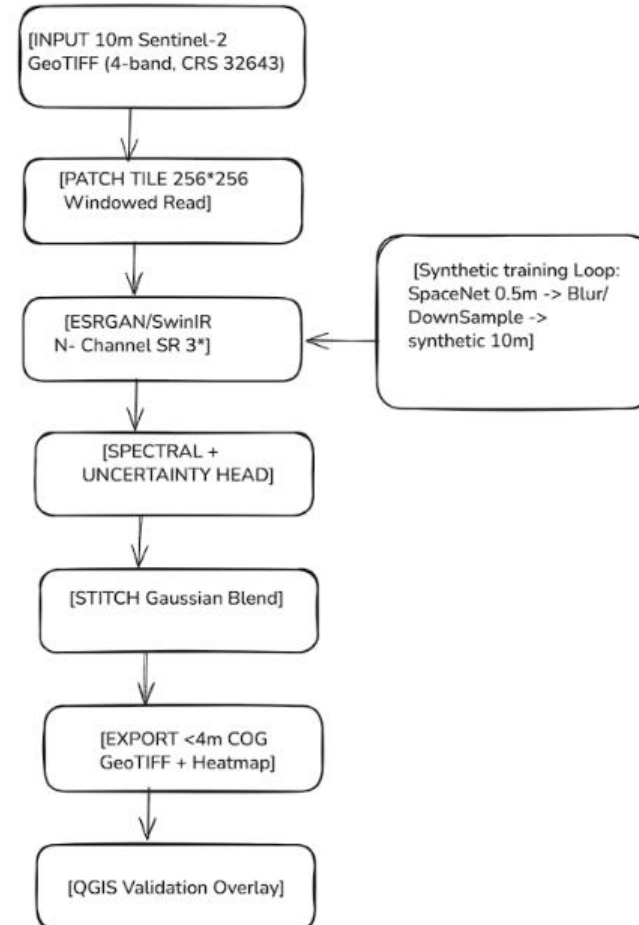


Synthetic degradation +
augmentations.
Loss: L1 + Perceptual (VGG) +
Spectral SAM.
Metrics: PSNR/SSIM/LPIPS/SAM

METHODOLOGY



PIPELINE IMPLEMENTATION & EXPORT FLOW



GitHub link: [Link](#)

**Above 60% of the
prototype is completed**

Direct Impact on Target Users

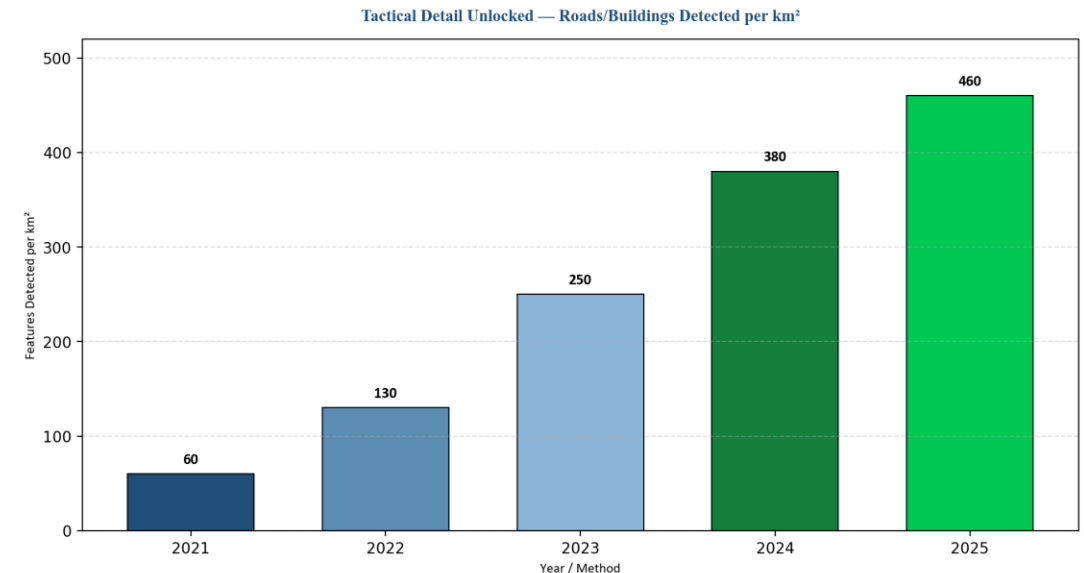
1. Defence & Intelligence (NTRO): Enables sovereign sub-4m tactical visibility for roads/buildings in hostile terrain (10m→3.3m in 3.1s).
2. Disaster Response (NDRF/SDRF/NDMA): Provides panchayat-level flood/damage mapping with QGIS COG overlay (faster than drone surveys).
3. Government & Urban Planning (ISRO): Ensures precise field-boundary & urban cadastre where (10m hides edges).
4. AgTech & Insurance: Delivers field delineation + post-disaster crop loss assessment using preserved NIR/NDVI no field visit.

Economic & Strategic Benefits

1. Eliminates \$1000s/scene commercial <1m buys
2. ₹111 Cr/year saved @₹5/km² (230M km²/yr),
3. TAM India Gov alone.
4. Superior spectral fidelity over bicubic/Topaz via N-channel + SAM.
5. Encourages ISRO-NTRO-AgTech PPP for scalable EO analytics.

Strategic Impact

1. Offers sovereign 3× super-resolution, resilient to hallucination via SAM + uncertainty heatmap.
2. Strengthens India's autonomy in defence, disaster mapping, and AgTech — free EO pipeline.
3. Impact: Contributes to Atmanirbhar Bharat by establishing indigenous <4m mapping for national security and governance.



Feasibility of an Idea



ANALYSIS OF FEASIBILITY

1. Stateless Flask API & tiled ML inference
2. Single Docker air-gapped for NTRO , no dependencies
3. Open Sentinel-2+SpaceNet, Colab Pro ₹2k vs Maxar ₹5-10L/10k km²



MARKET FEASIBILITY

1. Delivers sub-4m tactical resolution for NTRO, ISRO, NDMA, and AgTech without commercial feeds.
2. Zero recurring licensing fees projected
3. public savings of ~₹11+ Cr/year on satellite imagery.

POTENTIAL CHALLENGES & RISKS

1. Paired Data Scarcity: same day paired 10m – 3m satellite imagery
2. Hallucination: GAN fakes NDVI
3. Band Incompatibility: Uploads varying band counts(1,4,8)

STRATEGY FOR OVERCOMING THESE CHALLENGES

1. Synthetic PSF Degradation: Degrades 0.5m SpaceNet data With Gaussian Point Spread
2. Uncertainty Heatmaps: MC-Dropout T=10 viridis >0.6 red manual review
3. Dynamic Tiling Engine: Processes 256 x 256 tiles with 16x
4. Overlap and Gaussian blending to eliminate OOM risks

Key Research Papers

- **Real-ESRGAN:** Wang et al., ICCV 2021 (DOI: 10.1109/ICCVW54120).
- **DSen2 Model:** Lanaras et al., RSE 2018 (DOI: 10.1016/j.rse).
- **MC-Dropout:** Gal & Ghahramani, ICML 2016 (PMLR 48).
- **SAM Metrics:** Kruse et al., RSE 1993 (DOI: 10.1016/0034).

Open Datasets & Standards

- **SpaceNet Data:** High-res satellite catalog (spacenet.ai).
- **Sentinel-2 L2A:** Open multispectral data (Copernicus).
- **COG Standard:** OGC-compliant streaming raster format.

Core Frameworks

- **Rasterio:** Geospatial raster I/O processing (GDAL).
- PyTorch + torchvision Real-ESRGAN core/transforms.py
- **GeoTIFF.js:** WebGL browser georeferencing parser.
- Scikit-image / lpips / core/metrics.py